ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 2 [10 marks]

Q2(a) [2 marks]

At equilibrium the velocity should be zero. The torque and gravity torque must balance.

Q2(b) [2 marks]

$$A = [[0, 1], [-61.3125, -2.5]], B = [[0], [50]]$$

Q2(c) [4 marks]

The balance is 50(90-θss) = 2.4525 sin θss. Solving gives a small error of about 0.05°.

Q2(d) [2 marks]

The added gravity torque should cancel the load, so the steady-state error becomes zero.

Question 3 [10 marks]

Q3(a) [2 marks]

Finite differencing amplifies encoder noise, making the velocity estimate noisy.

Q3(b) [2 marks]

I use ωc=20 rad/s and α=ωcT/2=0.01. Then:

$$y[k] = 0.980198y[k-1] + 0.009901(x_k + x[k-1])$$

The DC gain is approximately 1.

Q3(c) [4 marks]

$$L = [18, 64]^T$$

Q3(d) [2 marks]

An observer can estimate velocity from position measurements. A filter can be safer because it smooths noise.

Question 4 [10 marks]

Q4(a) [2 marks]

The closed-loop poles are -2±2j. I use Nff=c/b=6 for unity DC gain.

Q4(b) [2 marks]

The nominal controller has unity DC gain, so θss=1 and the tracking error is zero even after the actuator gain changes.

Q4(c) [4 marks]

I obtain $s^3+(a+bk_2)s^2+(c+bk_1)s-bK_I$. Desired polynomial is $s^3+8s^2+24s+32$, so $k_2=6$, $k_1=18$, K_I=-32.

Q4(d) [2 marks]

Integral action removes steady-state error, although this only holds for the nominal values of b and c. Saturation can cause windup, which should be handled by anti-windup.